

(a) Move, don't regenerate



DINOv2 patches $\mathbf{Z}_0$
+ actions $\rightarrow \mathbf{Z}_{1:10}$

Existing WMs regenerate every patch



× drift × blur × compounding

ShiftWM moves what it has seen



$$\begin{aligned}\hat{\mathbf{Z}}_k = & (1-g)\mathbf{Z}_0 \text{ keep} \\ & + g\mathbf{T}_k \text{ move} \\ & + \mathbf{r}_k \text{ correct}\end{aligned}$$

✓ parallel from measured $\mathbf{Z}_0$ ✓ plug-in head

(b) One held-out DROID window

true $t+10$ (decoded)



ShiftWM (ours)



Direct



AR (recursive)



AR-TF (DINO-WM-style)



Persistence (copy t)



Each tile: that model's own forecast, one shared decoder.
Corner: feature error at $t+10$ (lower is better).

(c) Held-out results

beats Direct in 100/100 DROID horizon-motion bins

